

```
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
df=pd.read_csv("car_acceptability.csv")
df
```

	buying price	maintenance cost	number of doors	number of persons
0	vhigh	vhigh	2	2
1	vhigh	vhigh	2	2
2	vhigh	vhigh	2	2
3	vhigh	vhigh	2	2
4	vhigh	vhigh	2	2
...	...	...	...	...
1722	low	low	5	5
1723	low	low	5	5
1724	low	low	5	5
1725	low	low	5	5
1726	low	low	5	5

	lug_boot	safety	evaluation
0	small	med	unacc
1	small	high	unacc
2	med	low	unacc
3	med	med	unacc
4	med	high	unacc
...	...	...	...
1722	med	med	good
1723	med	high	vgood
1724	big	low	unacc
1725	big	med	good
1726	big	high	vgood

```
[1727 rows x 7 columns]
```

```
le = LabelEncoder()
```

```

df["buying price"]=le.fit_transform(df["buying price"])
df["maintenance cost"]=le.fit_transform(df["maintenance cost"])
df["lug_boot"]=le.fit_transform(df["lug_boot"])
df["safety"]=le.fit_transform(df["safety"])
df["evaluation"]=le.fit_transform(df["evaluation"])

```

```
df
```

	buying price	maintenance cost	number of doors	number of persons \
0	3	3	2	
2				
1	3	3	2	
2				
2	3	3	2	
2				
3	3	3	2	
2				
4	3	3	2	
2				
...	...	...	...	...
1722	1	1	5	
5				
1723	1	1	5	
5				
1724	1	1	5	
5				
1725	1	1	5	
5				
1726	1	1	5	
5				

	lug_boot	safety	evaluation
0	2	2	2
1	2	0	2
2	1	1	2
3	1	2	2
4	1	0	2
...	...	...	...
1722	1	2	1
1723	1	0	3
1724	0	1	2
1725	0	2	1
1726	0	0	3

```
[1727 rows x 7 columns]
```

```

X= df.drop("evaluation",axis=1)
Y=df["evaluation"]

```

```
X_train,X_test,Y_train,Y_test=train_test_split(X,Y,test_size=0.2,random_state=42)
```

```
rf=RandomForestClassifier(n_estimators=100,random_state=42)
rf.fit(X_train,Y_train)
```

```
RandomForestClassifier(random_state=42)
```

```
Y_pred=rf.predict(X_test)
```

```
print("Accuracy:",accuracy_score(Y_test,Y_pred))
```

```
print("\nConfusion Matrix:\n",confusion_matrix(Y_test,Y_pred))
```

```
Accuracy: 0.9624277456647399
```

```
Confusion Matrix:
```

```
[[ 72   1   3   1]
 [  2 10   0   3]
 [  1   0 236   0]
 [  2   0   0 15]]
```

```
import matplotlib.pyplot as plt
importance = rf.feature_importances_
plt.bar(X.columns,importance)
plt.xticks(rotation=45)
plt.xlabel("Features")
plt.ylabel("Importance")
plt.title("Random Forest Feature Importance")
plt.show()
```

